

投稿系统单独提交项: Graphical Abstract 说明文字 + Highlights

Running title

Delta Dependency for paralog prioritization

Graphical Abstract —Caption

Delta Dependency (DD) framework supported by exploratory analyses across five orthogonal layers. DD measures the shift in mean Chronos dependency of a paralog gene between driver wild-type and mutant cell lines (manuscript Eq. 1). Top: from DepMap 26Q1 data (1,208 cell lines; 66,595 HGNC paralog pairs; 12 curated, evidence-tiered pairs), DD nominates priority paralog-SL candidates without any training (AUROC 0.629 on the tiered curated set, pair-clustered 95% CI 0.253-0.933; 1.000 on the Tier A+B literature-derived benchmark, where both lineage-evaluable entries were the same pair). Bottom: exploratory support across genomic (23 solid tumor lineages; AUROC > 0.7 in 1 of 8 evaluable lineages on the primary frame, 6 of 12 on the relaxed sensitivity frame), proteomic (seven CPTAC cohorts, $n = 771$), pharmacologic (1,482 PRISM compounds; 633 selective associations), clinical-stratification (MSI status and mutation type), and structural (sequence features, domain architecture, PROTAC suitability) layers. Fully reproducible from public data with no GPU requirement.

Highlights (每条不超过 85 字符)

1. A single-subtraction metric (Delta Dependency) predicts paralog synthetic lethality
2. Interpretable composite (AUROC 0.831) matches best multi-feature classifier in CV
3. Protein-level paralog co-variation is detectable across seven CPTAC cohorts ($n = 771$)
4. Dependency-window analysis ranks established ARID1A-ARID1B top for selectivity
5. Every quantitative claim is recomputable from raw data with a single command